Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract and Executive Outline

Keywords: Prompt Caching; A/B Testing; Provider Routing; Cache Affinity; Latency; Throughput; Cost Attribution; gemini-3-flash-preview

Abstract

This report evaluates `google/gemini-3-flash-preview` on Infron and OpenRouter across cache reuse, observed cost, throughput, E2E latency, and Streaming TTFT under Prompt Caching workloads.

The main findings are: Infron leads Cache hit rate in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads Observed cost in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads Throughput in all routing modes; OpenRouter leads E2E latency in all routing modes; Infron leads Streaming TTFT in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie.

Overall, Infron's cross-mode strengths are throughput, while OpenRouter's cross-mode strengths are Streaming TTFT, E2E latency, observed cost, cache reuse. Platform choice should be driven by workload objective rather than a single headline metric.

Figure 0: Normalized Capability Radar

The five radar axes represent throughput, price, E2E latency, Streaming TTFT, and cache hit rate. Every metric is normalized to a 0–100 score, with farther outward meaning better.

Thick solid lines show platform-level contours; translucent lines and points show individual routing modes.

Conclusion Overview: Core Metric Winners by Routing Mode

Based on strict A/B paired samples. Blue represents Infron, orange represents OpenRouter; gold cells mark the winner for the routing objective.

Throughput	Observed cost	E2E latency	Streaming TTFT	Cache hit rate
Infron wins 4/4 Max advantage 2.79%, higher is better	OpenRouter wins 3/4 Max advantage 82.83%, lower is better	OpenRouter wins 4/4 Max advantage 38.55%, lower is better	OpenRouter wins 3/4 Max advantage 37.11%, lower is better	OpenRouter wins 3/4 Max advantage 37.03%, higher is better

Routing mode	Throughput objective	Cost objective	Latency objective	TTFT objective	Cache result
Throughput First throughput	Infron advantage 2.70%	OpenRouter advantage 9.29%	OpenRouter advantage 4.53%	OpenRouter advantage 2.72%	OpenRouter advantage 2.83%
Price First price	Infron advantage 2.34%	OpenRouter advantage 50.38%	OpenRouter advantage 13.17%	OpenRouter advantage 12.15%	OpenRouter advantage 37.03%
Latency First latency	Infron advantage 1.78%	OpenRouter advantage 17.89%	OpenRouter advantage 38.55%	OpenRouter advantage 37.11%	OpenRouter advantage 11.52%
TTFT First ttft	Infron advantage 2.79%	Infron advantage 82.83%	OpenRouter advantage 1.56%	Infron advantage 0.26%	Infron advantage 18.26%

Executive Outline

Dimension	Conclusion	Evidence
Controls	First/second `usage.prompt_tokens` deltas are limited to 50 tokens within each `sort/group/round` pair.	Methods and data quality
Cache reuse	Infron leads Cache hit rate in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie	Overall metrics and mechanism section
Observed cost	Infron leads Observed cost in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie	Overall metrics and provider drill-down
Performance	Infron leads Throughput in all routing modes; OpenRouter leads E2E latency in all routing modes; Infron leads Streaming TTFT in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie	Charts and statistical tests
Attribution boundary	Claims use observable response telemetry: usage, cost, TTFT, latency, provider fields, and cache tokens.	Provider/Route drill-down
Business meaning	Long-context, RAG-prefix, agent-tool, and high-frequency template workloads should evaluate cache rate, cost, first-token latency, and E2E latency together.	Discussion and conclusion

Routing-Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	**OpenRouter**	**OpenRouter**	**Infron**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (4/5 metrics)
Price First	Minimize request and token cost	**OpenRouter**	**OpenRouter**	**Infron**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (4/5 metrics)
Latency First	Minimize full-response waiting time	**OpenRouter**	**OpenRouter**	**Infron**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (4/5 metrics)
TTFT First	Minimize streaming first-token time	**Infron**	**Infron**	**Infron**	**OpenRouter**	**Infron**	Infron leads overall (4/5 metrics)

1. Introduction: Background, Questions, and Contributions

LLM inference-platform behavior is shaped by provider routing, prompt caching, streaming response handling, cost attribution, and fallback policy. This report treats the platform as an observable system and measures speed, cost, cache reuse, and first-token experience through strict A/B pairs.

1.1 Research Hypotheses

Hypothesis	Statement	Validation metric
H1	Stronger provider/cache affinity improves token-level cache hit rate for repeated stable prefixes.	Second-request cache-read tokens and token cache hit rate
H2	Higher cache hit rate can reduce observed cost, but does not necessarily reduce TTFT or E2E latency.	Observed cost, average TTFT, average request latency
H3	Different routing sorts change provider selection and produce different cost/throughput/latency frontiers.	Provider distribution, throughput, latency, cost

1.2 Contributions

- Uses response-returned `usage.prompt_tokens` as the input-token control while allowing small cross-platform accounting variance up to 50 tokens.
- Extends prompt-caching evaluation to cost, throughput, E2E latency, TTFT, provider distribution, reasoning telemetry, and paired statistical tests.
- Scopes every claim to observable response telemetry instead of private routing internals.

2. Experimental Design, Dataset, and Controls

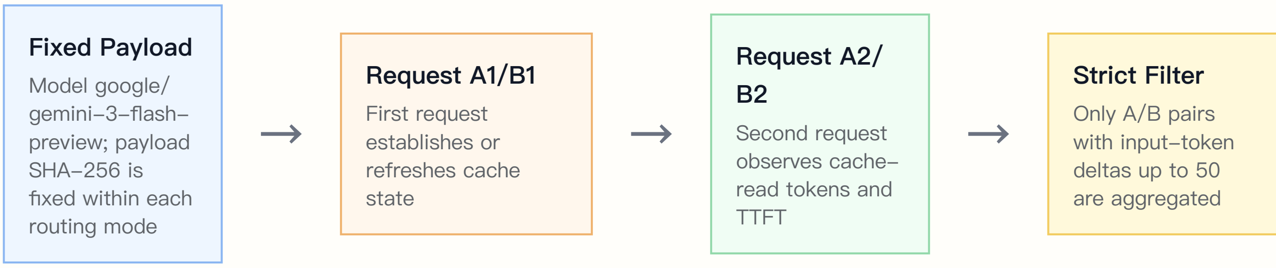
2.1 Dataset Construction

The dataset is `controlled_cache_probe`, covering 4 routing sorts, 2 platforms, 4 groups, and 50 rounds per group. Each round sends identical first/second Chat Completions requests; the second request observes cache-read tokens, TTFT, and E2E latency.

The built-in business templates represent stable long-context workloads such as RAG support, agent tool instructions, marketing automation, and code review.

2.2 Controlled Variables

Figure 1: Experimental Design and Strict A/B Pairing Filter



Controlled-variable rule: within the same `sort/group/round`, both platforms must have first/second `usage.prompt_tokens` deltas no greater than 50 tokens. Total Input Tokens are computed from response-returned usage, not local tokenizer estimates.

2.3 Metric Definitions

Metric	Definition	Direction
Call cache hit rate	Share of second requests with `cache_read_tokens > 0`	Higher is better
Token cache hit rate	Second-request cache-read tokens / second-request prompt tokens	Higher is better
Observed cost	Sum of first + second request usage/cost values	Lower is better
Throughput	Completion tokens / request latency seconds	Higher is better
E2E latency	Full response latency per request	Lower is better
TTFT	Streaming first-token arrival time	Lower is better
Reasoning telemetry	Reasoning tokens from response usage	Used to explain latency, throughput, and cost

3. Experimental Environment and Data Quality

Item	Configuration
Model	`google/gemini-3-flash-preview`
Provider model IDs	infron: `google/gemini-3-flash-preview`; openrouter: `google/gemini-3-flash-preview`
Platforms	Infron and OpenRouter
API protocol	`/v1/chat/completions`
Routing modes	Throughput First, Price First, Latency First, TTFT First
Groups	4
Rounds per group	50
Workers	24
Request mode	Streaming Chat Completions with TTFT collection
Reasoning / thinking control	No explicit reasoning/thinking parameter; model and platform defaults are preserved
Prompt length tiers	`short` ≈1500, `medium` ≈8000, `long` ≈32000
Excluded records	2

4. Results: Overall Metrics and Main Findings

Routing mode	Platform	Strict pairs	Total Input Tokens	Token cache hit rate	Observed cost	Throughput	E2E latency	Streaming TTFT
Throughput First	Infron	200	7397314	84.53%	$1.01312100	**4.14 tok/s**	2799.22 ms	2707.80 ms
Throughput First	**OpenRouter**	200	7397664	**86.92%**	**$0.92703585**	1.12 tok/s	**2677.88 ms**	**2636.09 ms**
Price First	Infron	200	7396914	63.73%	$1.23464300	**3.71 tok/s**	3052.44 ms	2981.14 ms
Price First	**OpenRouter**	200	7397418	**87.32%**	**$0.82103730**	1.11 tok/s	**2697.26 ms**	**2658.12 ms**
Latency First	Infron	200	7397106	77.67%	$0.97739800	**3.02 tok/s**	3828.60 ms	3747.77 ms
Latency First	**OpenRouter**	200	7397676	**86.62%**	**$0.82908675**	1.09 tok/s	**2763.24 ms**	**2733.50 ms**
TTFT First	**Infron**	199	7375738	**87.88%**	**$0.78488700**	**4.10 tok/s**	2819.77 ms	**2730.21 ms**
TTFT First	OpenRouter	199	7375762	74.31%	$1.43499695	1.08 tok/s	**2776.41 ms**	2737.43 ms

4.1 Tail Latency and Statistical Tests

Tail percentiles expose risk that averages hide.

Routing mode	Platform	P50 latency	P95 latency	P99 latency	P50 TTFT	P95 TTFT	P99 TTFT
Throughput First	Infron	2679.33 ms	4424.36 ms	5260.37 ms	**2580.68 ms**	4284.46 ms	5178.37 ms
Throughput First	**OpenRouter**	**2678.93 ms**	**3955.32 ms**	**4465.37 ms**	2630.48 ms	**3916.57 ms**	**4457.56 ms**
Price First	Infron	2798.90 ms	5305.44 ms	7888.42 ms	2709.57 ms	5194.32 ms	7762.28 ms
Price First	**OpenRouter**	**2706.74 ms**	**4010.61 ms**	**5045.69 ms**	**2676.98 ms**	**3970.16 ms**	**4988.11 ms**
Latency First	Infron	2944.29 ms	9579.16 ms	11308.45 ms	2855.95 ms	9456.74 ms	11226.69 ms
Latency First	**OpenRouter**	**2711.76 ms**	**3973.23 ms**	**4734.37 ms**	**2677.92 ms**	**3935.48 ms**	**4708.08 ms**
TTFT First	**Infron**	2741.43 ms	4319.60 ms	4931.79 ms	**2662.26 ms**	4197.65 ms	4812.57 ms
TTFT First	OpenRouter	**2716.23 ms**	**4096.98 ms**	**4738.42 ms**	2676.31 ms	**4056.45 ms**	**4706.44 ms**

Mean deltas use bootstrap 95% CIs; p-values use paired sign-flip permutation tests.

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Throughput First	Latency: OpenRouter − Infron	**−242.68 ms**	−376.53 ms to −104.63 ms	<0.001	200	Positive means lower Infron latency
Throughput First	TTFT: OpenRouter − Infron	**−143.43 ms**	−275.00 ms to −6.56 ms	0.0405	200	Positive means lower Infron TTFT
Throughput First	Throughput: Infron − OpenRouter	**3.4043 tok/s**	3.1786 tok/s to 3.6218 tok/s	<0.001	200	Positive means higher Infron throughput
Throughput First	Cost: OpenRouter − Infron	**$−0.00043043**	$−0.00118760 to $0.00019729	0.2514	200	Positive means lower Infron cost
Throughput First	Token Cache Hit: Infron − OpenRouter	**0.12 pp**	−2.07 pp to 2.30 pp	0.9865	200	Positive means higher Infron cache hit
Price First	Latency: OpenRouter − Infron	**−710.36 ms**	−938.53 ms to −480.55 ms	<0.001	200	Positive means lower Infron latency
Price First	TTFT: OpenRouter − Infron	**−646.03 ms**	−872.33 ms to −417.64 ms	<0.001	200	Positive means lower Infron TTFT
Price First	Throughput: Infron − OpenRouter	**2.9611 tok/s**	2.7695 tok/s to 3.1621 tok/s	<0.001	200	Positive means higher Infron throughput
Price First	Cost: OpenRouter − Infron	**$−0.00206803**	$−0.00302432 to $−0.00121534	<0.001	200	Positive means lower Infron cost
Price First	Token Cache Hit: Infron − OpenRouter	**−13.49 pp**	−18.42 pp to −9.03 pp	<0.001	200	Positive means higher Infron cache hit
Latency First	Latency: OpenRouter − Infron	**−2130.72 ms**	−2647.60 ms to −1627.08 ms	<0.001	200	Positive means lower Infron latency
Latency First	TTFT: OpenRouter − Infron	**−2028.55 ms**	−2547.48 ms to −1524.20 ms	<0.001	200	Positive means lower Infron TTFT

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Latency First	Throughput: Infron – OpenRouter	**2.6086 tok/s**	2.3734 tok/s to 2.8461 tok/s	<0.001	200	Positive means higher Infron throughput
Latency First	Cost: OpenRouter – Infron	**$–0.00074156**	$–0.00125460 to $–0.00025523	0.0020	200	Positive means lower Infron cost
Latency First	Token Cache Hit: Infron – OpenRouter	**–8.31 pp**	–11.81 pp to –5.03 pp	<0.001	200	Positive means higher Infron cache hit
TTFT First	Latency: OpenRouter – Infron	**–86.73 ms**	–221.65 ms to 67.44 ms	0.2387	199	Positive means lower Infron latency
TTFT First	TTFT: OpenRouter – Infron	**14.44 ms**	–120.69 ms to 163.43 ms	0.8338	199	Positive means lower Infron TTFT
TTFT First	Throughput: Infron – OpenRouter	**3.3119 tok/s**	3.1220 tok/s to 3.5107 tok/s	<0.001	199	Positive means higher Infron throughput
TTFT First	Cost: OpenRouter – Infron	**$0.00326688**	$0.00223199 to $0.00443892	<0.001	199	Positive means lower Infron cost
TTFT First	Token Cache Hit: Infron – OpenRouter	**8.53 pp**	5.30 pp to 12.11 pp	<0.001	199	Positive means higher Infron cache hit

4.2 Reasoning / Thinking Control Check

This run does not explicitly set reasoning/thinking parameters and keeps model/platform defaults; this table records reasoning telemetry under default behavior.

Routing mode	Platform	Reasoning tokens	Avg reasoning tokens/request	Reasoning requests	Avg first reasoning token	Avg TTFT	Avg E2E latency
Throughput First	Infron	4513	11.2825	372	**0.00 ms**	2707.80 ms	2799.22 ms
Throughput First	**OpenRouter**	**0**	**0.0000**	**0**	2666.95 ms	**2636.09 ms**	**2677.88 ms**
Price First	Infron	4359	10.8975	367	**0.00 ms**	2981.14 ms	3052.44 ms
Price First	**OpenRouter**	**0**	**0.0000**	**0**	2680.65 ms	**2658.12 ms**	**2697.26 ms**
Latency First	Infron	4498	11.2450	374	**0.00 ms**	3747.77 ms	3828.60 ms
Latency First	**OpenRouter**	**0**	**0.0000**	**0**	2748.99 ms	**2733.50 ms**	**2763.24 ms**
TTFT First	**Infron**	4473	11.2387	369	**0.00 ms**	**2730.21 ms**	2819.77 ms
TTFT First	OpenRouter	**0**	**0.0000**	**0**	2759.65 ms	2737.43 ms	**2776.41 ms**

4.3 API Protocol Compatibility Matrix

This run uses `/v1/chat/completions`; this table records success response, usage, cost, and cache telemetry coverage for both platforms under that protocol.

API protocol	Endpoint	Platform	Pairs	Requests	Success rate	Usage coverage	Token usage coverage	Cost coverage	Cache telemetry coverage
`chat_completions`	`/v1/chat/completions`	Infron	800	1600	**100.00%**	**100.00%**	**100.00%**	**100.00%**	**100.00%**
`chat_completions`	`/v1/chat/completions`	OpenRouter	800	1600	99.94%	**100.00%**	**100.00%**	**100.00%**	**100.00%**

5. Result Visualizations by Routing Mode

The following charts are driven by the strict-paired summary data and present routing-mode, platform, cost, cache, E2E latency, Streaming TTFT, and upstream provider differences.

5.1 Core Metric Chart Overview

The charts below are generated from the same strict-paired summary data.

Figure 3-E2: Normalized capability contour

Five metrics are normalized to 0–100, where higher is better.

Figure 3-E3: Cost-cache efficiency plane

X-axis is token cache hit rate; Y-axis is observed total cost.

Figure 3-E1: Routing-mode core metric comparison

Infron and OpenRouter are shown side by side by side by routing mode.

Figure 3-E4: E2E latency and Streaming TTFT

Solid lines show E2E latency; dashed lines show Streaming TTFT.

Figure 3-E5: Upstream provider distribution

Provider shares explain cache-domain and performance differences.

Routing-Mode Drill-Down Charts

The horizontal axis shows relative advantage: blue to the right indicates Infron advantage, orange to the left indicates OpenRouter advantage.

Figure 4–E1: Throughput First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E2: Price First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E3: Latency First core metric comparison

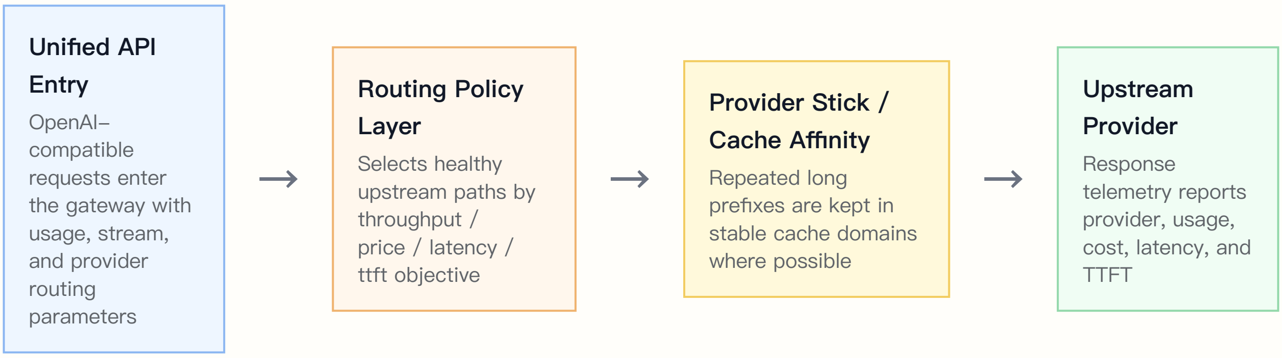
Direction and magnitude of relative advantage across five metrics.

Figure 4–E4: TTFT First core metric comparison

Direction and magnitude of relative advantage across five metrics.

6. Infron Technical Architecture and Cache/Cost Mechanism

Figure 12: Infron Multi-Provider Routing and Cache Control Plane



6.1 Multi-Provider Routing and Observable Control Plane

Requests enter a unified API, and the routing layer selects healthy upstream paths according to throughput, price, latency, or ttft objectives. Whether stable long prefixes stay in the same cache domain directly affects second-request cache-read tokens.

6.2 Provider Stick and Cache Affinity

Provider stick is cache affinity, not a permanent provider lock. Its goal is to reduce cache-domain fragmentation while respecting health and SLA constraints.

6.3 Cost-Control Path

Mechanism	Cache-rate effect	Cost effect	Observable signal
Stable prefix detection	Repeated prefixes are more likely to hit cache	Reduces repeated prefill cost	Payload SHA-256 and second-request cache-read tokens
Provider stick / cache affinity	Reduces cross-domain cache fragmentation	Avoids repeated cache warm-up	Provider distribution and token cache hit rate
Health checks and fallback	Protects availability while sometimes sacrificing cache	Reduces failure cost	HTTP status, provider distribution, tail latency
Cost-aware routing	Prefers lower-cost paths under constraints	Reduces total and per-round cost	Observed cost, cost breakdown coverage, cache-read tokens

7. Provider/Route Drill-Down

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	Infron	400	400	`google-ai-studio` 400

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	OpenRouter	400	400	`Google AI Studio` 375, `Google` 25
Price First	Infron	400	400	`google-vertex` 400
Price First	OpenRouter	400	400	`Google AI Studio` 252, `Google` 148
Latency First	Infron	400	400	`google-vertex` 208, `google-ai-studio` 192
Latency First	OpenRouter	400	400	`Google AI Studio` 381, `Google` 19
TTFT First	Infron	398	398	`google-ai-studio` 398
TTFT First	OpenRouter	398	398	`Google AI Studio` 211, `Google` 187

Upstream Provider Detail Distribution

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens	Rea toke
Throughput First	Infron	`google-ai-studio`	400	100.00%	200/200	200	2707.80 ms	2799.22 ms	7397314	4640	451
Throughput First	OpenRouter	`Google AI Studio`	375	93.75%	186/189	198	2650.34 ms	2692.98 ms	7030700	1125	0
Throughput First	OpenRouter	`Google`	25	6.25%	14/11	23	2422.36 ms	2451.45 ms	366964	75	0
Price First	Infron	`google-vertex`	400	100.00%	200/200	200	2981.14 ms	3052.44 ms	7396914	4530	435
Price First	OpenRouter	`Google AI Studio`	252	63.00%	125/127	129	3088.90 ms	3132.43 ms	6872712	756	0
Price First	OpenRouter	`Google`	148	37.00%	75/73	77	1924.63 ms	1956.28 ms	524706	444	0
Latency First	Infron	`google-vertex`	208	52.00%	104/104	116	4690.35 ms	4762.75 ms	3945758	2375	229
Latency First	Infron	`google-ai-studio`	192	48.00%	96/96	108	2726.65 ms	2816.60 ms	3451348	2252	220
Latency First	OpenRouter	`Google AI Studio`	381	95.25%	192/189	195	2738.17 ms	2767.93 ms	7084040	1143	0
Latency First	OpenRouter	`Google`	19	4.75%	8/11	14	2639.78 ms	2669.24 ms	313636	57	0

Routing mode	Platform	Upstream provider	Requests	Share	first/ second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens	Rea tok
TTFT First	Infron	`google-ai-studio`	398	100.00%	199/199	199	2730.21 ms	2819.77 ms	7375738	4603	447
TTFT First	OpenRouter	`Google AI Studio`	211	53.02%	107/104	108	2734.78 ms	2765.80 ms	3756494	633	0
TTFT First	OpenRouter	`Google`	187	46.98%	92/95	96	2740.41 ms	2788.38 ms	3619268	561	0

7.1 Cache-Rate and Cost Divergence Drill-Down

This table combines cache, cost, provider distribution, and reasoning telemetry to explain routing-mode differences.

Routing mode	Cache-hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
Throughput First	-2.39 pp	1.09x	`google-ai-studio` 100.00%	**`Google AI Studio` 93.75%**	+4513	OpenRouter has higher cache and lower cost; provider/ cache-domain mix is the main signal
Price First	-23.60 pp	1.50x	`google-vertex` 100.00%	**`Google AI Studio` 63.00%**	+4359	OpenRouter has higher cache and lower cost; provider/ cache-domain mix is the main signal
Latency First	-8.95 pp	1.18x	`google-vertex` 52.00%	**`Google AI Studio` 95.25%**	+4498	OpenRouter has higher cache and lower cost; provider/ cache-domain mix is the main signal
TTFT First	**+13.57 pp**	**0.55x**	**`google-ai-studio` 100.00%**	`Google AI Studio` 53.02%	+4473	Infron leads both cache and cost

8. Stratified Results: Prompt-Length Cache Performance

This section aggregates second-request cache-read tokens, token-level cache hit rate, observed cost, E2E latency, and Streaming TTFT by prompt-length tier. Bold cells mark the advantaged side within each tier.

Prompt-Length Tier Overview

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
`short`	1500	Infron	**268**	554343	**0**	**0.00%**	**$0.52029200**	2194.54 ms	2138.04 ms
`short`	1500	OpenRouter	**268**	**554496**	**0**	**0.00%**	$0.55932500	**1916.73 ms**	**1885.15 ms**
`medium`	8000	Infron	**267**	2880246	1936455	67.23%	**$1.04314200**	3127.68 ms	3052.55 ms
`medium`	8000	OpenRouter	**267**	**2880529**	**2051351**	**71.21%**	$1.09885910	**2763.20 ms**	**2727.90 ms**
`long`	32000	Infron	**264**	11348947	9660348	85.12%	$2.44661500	4068.03 ms	3949.35 ms
`long`	32000	OpenRouter	**264**	**11349234**	**10337850**	**91.09%**	**$2.35397275**	**3517.89 ms**	**3472.42 ms**

Prompt Length x Routing Mode Cache Hit Rate

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
`short`	Throughput First	**0.00%**	**0.00%**	tie
`short`	Price First	**0.00%**	**0.00%**	tie
`short`	Latency First	**0.00%**	**0.00%**	tie
`short`	TTFT First	**0.00%**	**0.00%**	tie
`medium`	Throughput First	**75.74%**	71.17%	**Infron**
`medium`	Price First	59.91%	**73.22%**	**OpenRouter**
`medium`	Latency First	57.66%	**75.41%**	**OpenRouter**
`medium`	TTFT First	**75.74%**	64.96%	**Infron**
`long`	Throughput First	90.89%	**95.18%**	**OpenRouter**
`long`	Price First	67.81%	**95.18%**	**OpenRouter**
`long`	Latency First	86.56%	**93.71%**	**OpenRouter**
`long`	TTFT First	**95.22%**	80.29%	**Infron**

9. Stratified Results: Group-Level Stability

Throughput First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	74.04%	$0.42006300	4194.73 ms	4068.78 ms
Infron	2	50	50	**88.12%**	**$0.19941600**	4476.24 ms	4346.78 ms
Infron	3	50	50	**88.07%**	**$0.19810300**	4450.53 ms	4374.59 ms
Infron	4	50	50	**87.58%**	**$0.19553900**	4401.49 ms	4271.98 ms
OpenRouter	1	50	50	**85.76%**	**$0.26666735**	**4063.60 ms**	**4020.12 ms**
OpenRouter	2	50	50	87.23%	$0.22678760	**3871.40 ms**	**3851.67 ms**
OpenRouter	3	50	50	88.04%	$0.21417565	**3876.74 ms**	**3866.94 ms**
OpenRouter	4	50	50	86.58%	$0.21940525	**3749.42 ms**	**3709.10 ms**

Price First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	47.85%	$0.43105600	5400.33 ms	5316.99 ms
Infron	2	50	50	64.79%	$0.31797800	5408.31 ms	5315.10 ms
Infron	3	50	50	61.91%	$0.27840300	4550.87 ms	4452.77 ms
Infron	4	50	50	80.36%	$0.20720600	5324.10 ms	5284.51 ms
OpenRouter	1	50	50	**87.56%**	**$0.19668510**	**3758.50 ms**	**3727.86 ms**
OpenRouter	2	50	50	**87.18%**	**$0.20489900**	**3805.17 ms**	**3752.18 ms**
OpenRouter	3	50	50	**87.99%**	**$0.22171730**	**4272.67 ms**	**4199.24 ms**
OpenRouter	4	50	50	**86.53%**	**$0.19773590**	**4145.74 ms**	**4104.25 ms**

Latency First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	85.78%	$0.19940200	8791.63 ms	8760.05 ms
Infron	2	50	50	58.75%	$0.31247400	10021.31 ms	9946.22 ms
Infron	3	50	50	79.36%	$0.27000700	9757.91 ms	9564.77 ms
Infron	4	50	50	**87.58%**	**$0.19551500**	**4043.35 ms**	**3873.91 ms**
OpenRouter	1	50	50	**87.50%**	**$0.19694535**	**3966.10 ms**	**3935.48 ms**
OpenRouter	2	50	50	**88.04%**	**$0.19758260**	**3860.03 ms**	**3827.32 ms**
OpenRouter	3	50	50	**83.61%**	**$0.21844765**	**3978.71 ms**	**3924.27 ms**

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
OpenRouter	4	50	50	87.37%	$0.21611115	4057.08 ms	4017.68 ms

TTFT First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	**87.58%**	**$0.19536800**	**3928.59 ms**	**3795.68 ms**
Infron	2	50	50	**88.12%**	**$0.19941300**	4471.03 ms	4353.37 ms
Infron	3	50	50	**88.07%**	**$0.19810000**	4297.33 ms	4150.03 ms
Infron	4	49	49	**87.73%**	**$0.19200600**	4124.18 ms	4008.31 ms
OpenRouter	1	50	50	82.14%	$0.21513325	4257.46 ms	4229.30 ms
OpenRouter	2	50	50	83.72%	$0.21599975	**3759.37 ms**	**3756.22 ms**
OpenRouter	3	50	50	66.73%	$0.56662125	**4130.57 ms**	**4116.54 ms**
OpenRouter	4	49	49	64.36%	$0.43724270	**3953.41 ms**	**3901.82 ms**

10. Discussion: Business Value, Boundaries, and Engineering Implications

Business decisions should not rely on one metric. Stable long-context and high-frequency template workloads should prioritize cache rate and cost; realtime interaction must constrain TTFT and E2E latency; batch processing often prioritizes throughput and failure cost.

Routing mode	Business objective	Observed result	Scenarios	Caveat
Throughput First	Maximize output capacity per unit time	OpenRouter leads overall (4/5 metrics)	Batch generation, offline summaries, backend processing	First-token and E2E response are faster; check cache and cost
Price First	Minimize request and token cost	OpenRouter leads overall (4/5 metrics)	High-frequency templates, support automation, marketing, RAG prefixes	First-token and E2E response are faster; check cache and cost
Latency First	Minimize full-response waiting time	OpenRouter leads overall (4/5 metrics)	Online chat, agent chains, IDE/writing assistants, realtime tools	First-token and E2E response are faster; check cache and cost
TTFT First	Minimize streaming first-token time	Infron leads overall (4/5 metrics)	Streaming chat, realtime copilots, first-screen feedback	Cache and cost are stronger; still check speed SLA

11. Conclusion

Routing-Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	**OpenRouter**	**OpenRouter**	**Infron**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (4/5 metrics)
Price First	Minimize request and token cost	**OpenRouter**	**OpenRouter**	**Infron**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (4/5 metrics)
Latency First	Minimize full-response waiting time	**OpenRouter**	**OpenRouter**	**Infron**	**OpenRouter**	**OpenRouter**	OpenRouter leads overall (4/5 metrics)
TTFT First	Minimize streaming first-token time	**Infron**	**Infron**	**Infron**	**OpenRouter**	**Infron**	Infron leads overall (4/5 metrics)

12. Limitations, Missing Data, and Next Experiments

Limitation	Impact	Next step	Current handling
Full routing trace	Cannot prove every provider choice, fallback, and retry path hop by hop	Add provider routing trace, decision logs, and fallback reasons	Use only returned provider fields and provider distribution
Longer time window	4x50 observes short-window stability but not day-level drift	Add soak tests and repeated windows	Scope conclusions to this run
Production corpus	Built-in templates do not cover every workload distribution	Use sanitized production-stratified corpora	Discuss representative long-context templates only
Cost-field consistency	Cost coverage and semantics may differ by platform	Reconcile with billing and provider cost breakdown	Use only explicitly returned cost fields

13. Reproducibility Appendix

Artifact	Path
Summary	`export/gemini3_flash_preview_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completion summary.json`
Paired dataset	`export/gemini3_flash_preview_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completion benchmark_pairs.csv`

Artifact	Path
Request-level dataset	`export/gemini3_flash_preview_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_requests.jsonl`
Filtered structured records	`export/gemini3_flash_preview_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions records.json`
Excluded-record audit	`export/gemini3_flash_preview_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions records_excluded.json`
Test source	`tests/test_rerun_routing_sort_cache_cost_ab.py`
Benchmark runner source	`scripts/rerun_routing_sort_cache_cost_ab.py`
HTML report renderer source	`scripts/render_glm52_deepseek_style_report.py`
Dataset reference	`business_representative` built-in representative business templates; request-level export is `benchmark_requests`